Industry Classification of GitHub Repositories Using the North American Industry Classification System (NAICS)

Kevin Xu, Alexander Quispe
khxu@github.com, alexanderquispe@github.com

May 22, 2026

What is the North American Industry Classification System (NAICS)?

- Standard system for classifying **business establishments** by industry.
- Jointly developed by the U.S., Canada, and Mexico.
- Used by governments, researchers, and businesses for:
  - Economic analysis and statistics
  - Policy and regulation
  - Industry benchmarking
- Hierarchical structure:
  - 2-digit codes = 20 broad sectors (e.g., **11: Agriculture**)
  - 3-6 digit codes = more specific sub-industries (e.g., **Soybean Farming, Oilseed Farming , Dry Pea and Bean Farming , Wheat Farming...**)

NAICS 20 Broad Sectors

NAICS Code Drill-Down Table

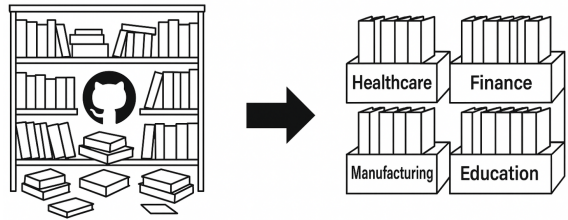
Click on any NAICS Code to see the Top Business Profiles within that Industry.

Code	Sector Title	Number of US Entities
11	Agriculture, Forestry, Fishing and Hunting	367,959
21	Mining	33,339
22	Utilities	52,270
23	Construction	1,512,763
31-33	Manufacturing	660,640
42	Wholesale Trade	698,477
44-45	Retail Trade	1,870,617
48-49	Transportation and Warehousing	711,582
51	Information	370,463
52	Finance and Insurance	771,419
	Total US Business Entities	7,049,529

Code	Sector Title	Number of US Entities
53	Real Estate Rental and Leasing	926,476
54	Professional, Scientific, and Technical Services	2,489,746
55	Management of Companies and Enterprises	93,116
56	Administrative and Support and Waste... Services	1,553,879
61	Educational Services	430,343
62	Health Care and Social Assistance	1,695,931
71	Arts, Entertainment, and Recreation	385,880
72	Accommodation and Food Services	931,927
81	Other Services (except Public Administration)	1,955,493
92	Public Administration	256,211
	Total US Business Entities	10,719,002

The Problem

- GitHub hosts millions of repositories, but they lack clear **industry classification**.
- Policymakers, companies, and researchers need to know **where innovation is happening**.
- Without structure, GitHub is like a **library without categories**.
- **Our goal: build a system to classify repositories into industries (e.g., Healthcare, Finance, Manufacturing).**



Roadmap

1. **Creating the Training Dataset**
   From raw GitHub repositories to industry-labeled data.
2. **Models Used for Fine Tuning**
   Comparing state-of-the-art language models for classification.
3. **Inference**
   Predicting the industry of new repositories.

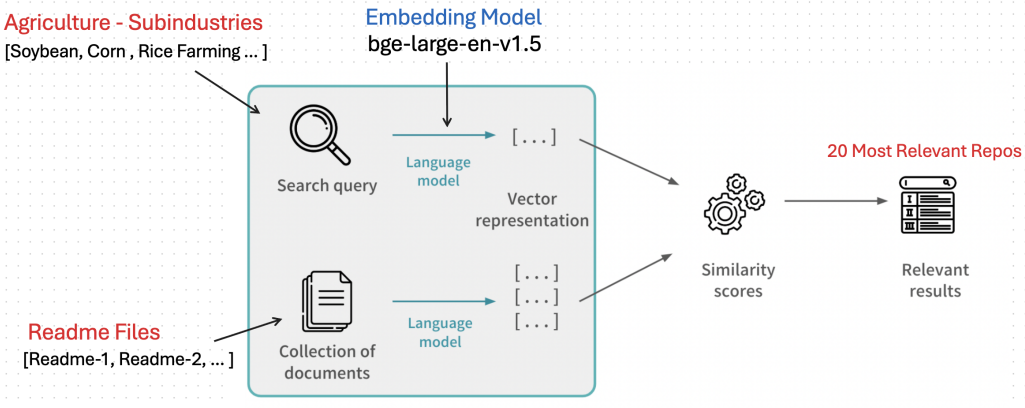
Step 1: Creating the Training Dataset - Semantic Match

- **Inputs:**
  - **Industry Keywords (NAICS subindustries)** e.g., for Agriculture we have “Soybean Farming, Oilseed Farming , Dry Pea and Bean Farming , Wheat Farming...”
  - **511k GitHub Repositories from USA** - Repository README files + metadata (name, description, topics)

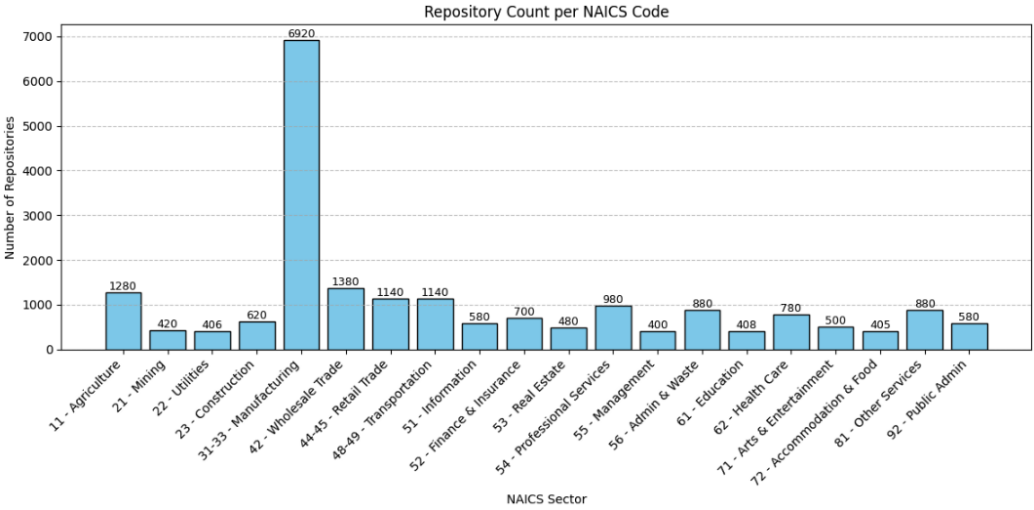
- **Process:**
  - Convert both industries and repositories into vector embeddings.
  - Measure similarity between subindustry keywords and repositories.
  - Retrieve top 20 candidates per subindustry.

- **Outputs:**
  - Candidate repositories matched to industries.
  - Around **10, 983 samples** selected from 511k repositories.

Step 1: Creating the Training Dataset - Semantic Match Workflow



Step 1: Creating the Training Dataset-Semantic Match Results



Step 1: Creating the Training Dataset - Validation with GPT-4 Agents

Goal: Confirm semantic matches with higher precision.

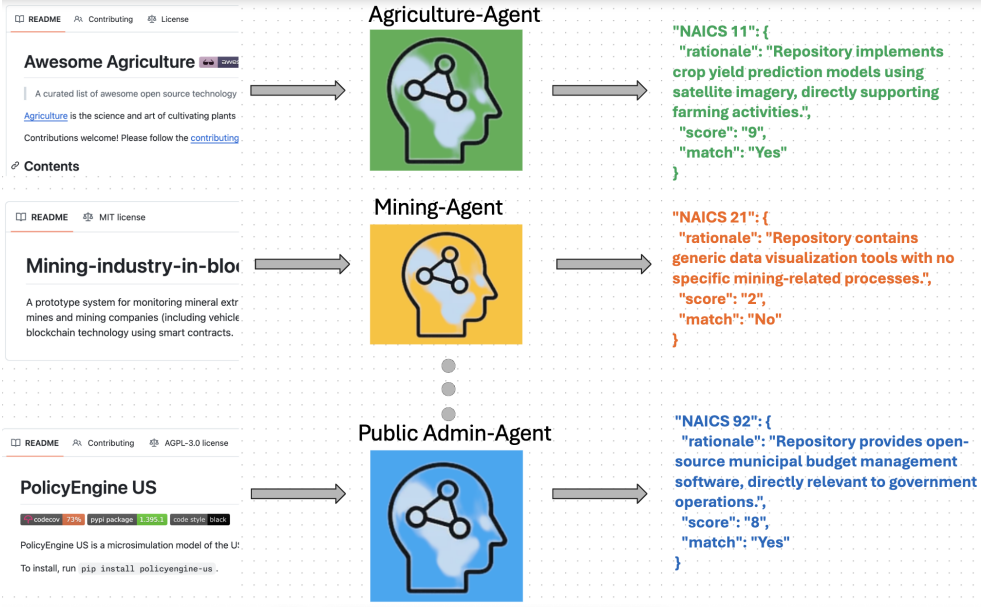
Process

- Input: README content + target NAICS sector description.
- GPT-4 Agent applies a classification framework:
  - Industry-specific software
  - Sector-relevant functionality
  - Domain applications
  - Sector-specific data/research
- Decision Standards:
  - **Include (Yes):** clear sector alignment
  - **Exclude (No):** generic or tangential

Output

- Structured JSON: `"NAICS 11": {"rationale": "...", "score": "1-10", "match": "Yes/No"}`
- Filter: keep only repositories with score ≥ 8
- Final dataset: $\sim$2,529 high-confidence repositories

Step 1: Creating the Training Dataset - GPT4 NAICS Agents



Step 1: Creating the Training Dataset - Validation results

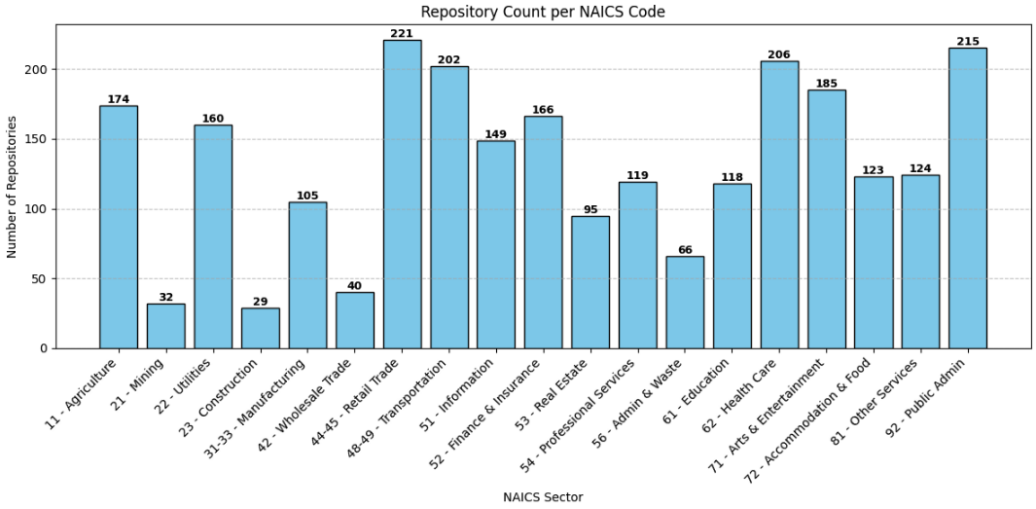


Figure 2: GPT4.1 Classification for USA Repos

Step 1: Creating the Training Dataset - Expanding Beyond U.S. Data

- Added repositories from the **European Union** and **Australia**, alongside U.S. data.
- **Australia** was included to strengthen coverage in the **Mining** sector.
- **Total repositories:** 6,588
- **NAICS categories:** 19 business sectors (no data available for Category 55: Management of Companies and Enterprises).

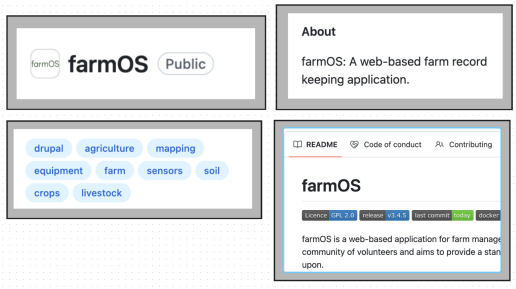
Step 2: Fine-Tuning Process

Dataset Composition

- **Total repositories:** 6,588 projects
- **NAICS categories:** 19 industry codes
- **Split:** 70% training (4,611)/ 10% validation (658) / 20% testing (1,319)

Text Features Used

- Repository **name** and **description**
- **Topic tags** (keywords given by developers)
- **README content**
- Combined input: ∼340–390 words on average
- Capped at 512 tokens



Step 2: Fine-Tuning Process, Why Transformers for Repository Classification?

Why Transformers?

- State-of-the-art for **natural language understanding**.
- Can handle long documents like **README files** and metadata.
- Capture both **local details** (specific keywords) and **global context** (overall repository purpose).

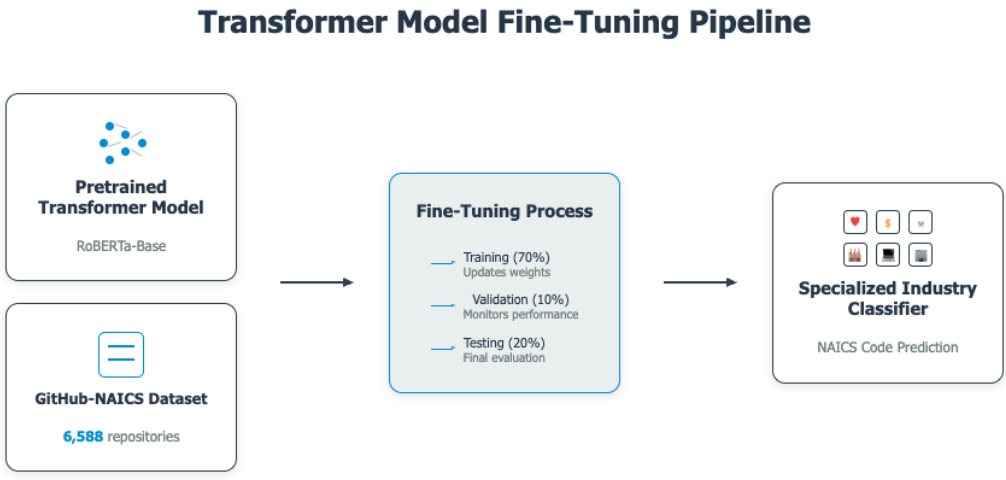
Model Capacity

- Base models: $\sim$100M parameters (faster, smaller).
- Large models: 300–400M parameters (more accurate, higher computational cost).

Fine-Tuning

- Start from a **pretrained Transformer** (trained on massive text corpora).
- Adjust weights with our labeled GitHub-NAICS dataset so the model learns **industry-specific classification**.
- This avoids training from scratch, saving time and resources.

Step 2: Fine-Tuning Workflow



Step 2: Fine-Tuning, Results from Testing Data

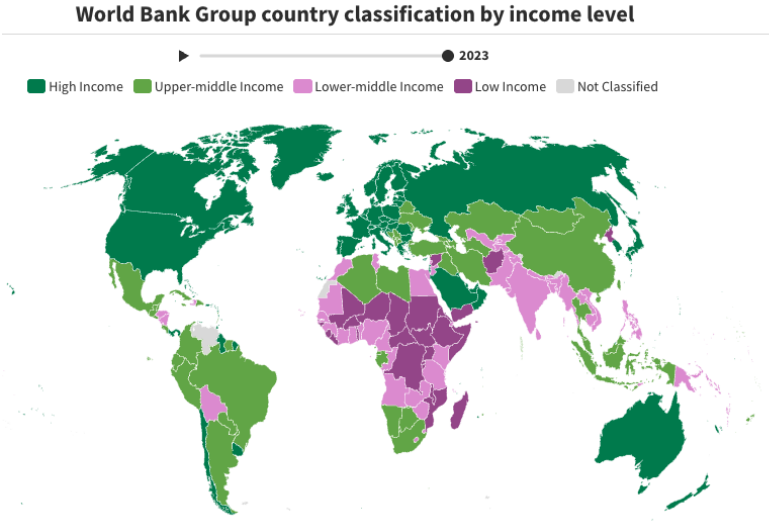
Model	Parameters	F1 Score	Accuracy	Precision	Recall
RoBERTa-base	125M	84.26%	84.84%	84.70%	84.84%
RoBERTa-large	355M	**86.45%**	**86.35%**	**86.68%**	**86.35%**
ModernBERT-base	139M	84.63%	84.85%	84.86%	84.85%
ModernBERT-large	395M	84.16%	84.23%	84.44%	84.23%
DeBERTa-v3-base	183M	85.68%	85.82%	86.16%	85.82%
DeBERTa-v3-large	400M	85.07%	85.22%	85.41%	85.22%

Table 1: Performance of transformer models by size (base vs. large).

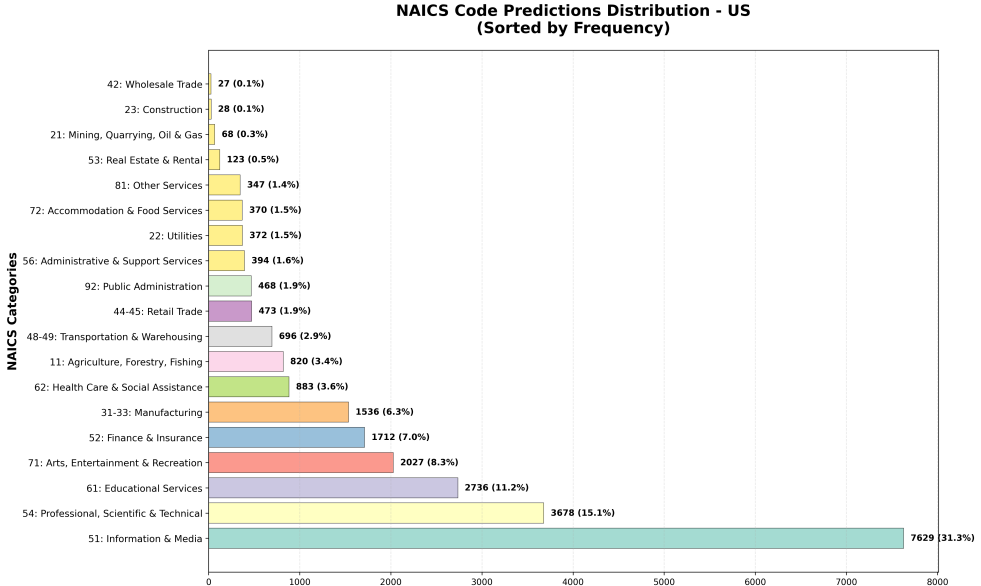
Conclusions:

- **RoBERTa-large** achieved the best overall performance (F1 = 86.45%), showing that scaling improves accuracy.
- **Model family matters:** DeBERTa-v3 and ModernBERT provided competitive results, but RoBERTa remained strongest on this dataset.

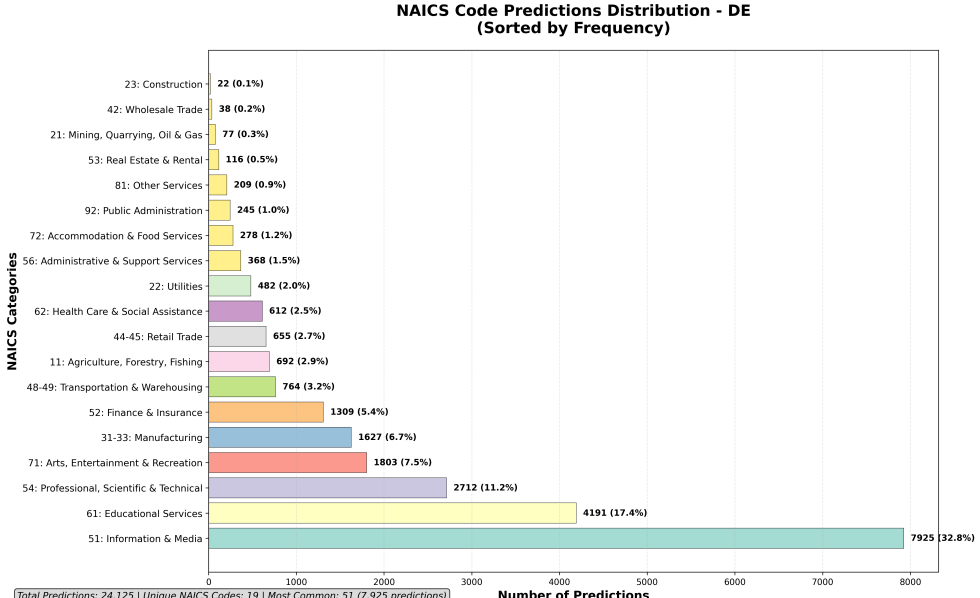
World Bank Group country classification by income level



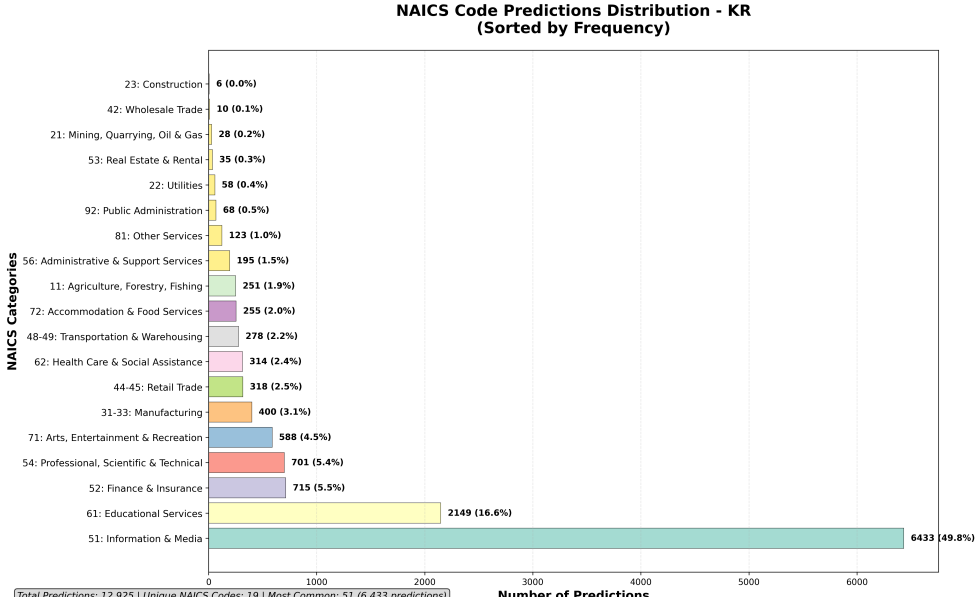
Step 3: Inference, High-Income Economies - USA



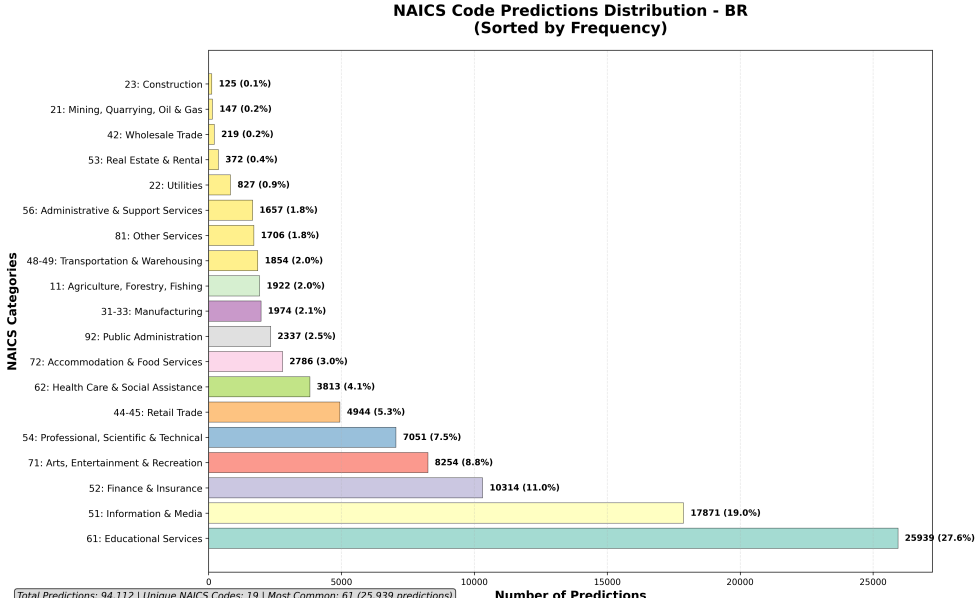
Step 3: Inference, High-Income Economies - Germany



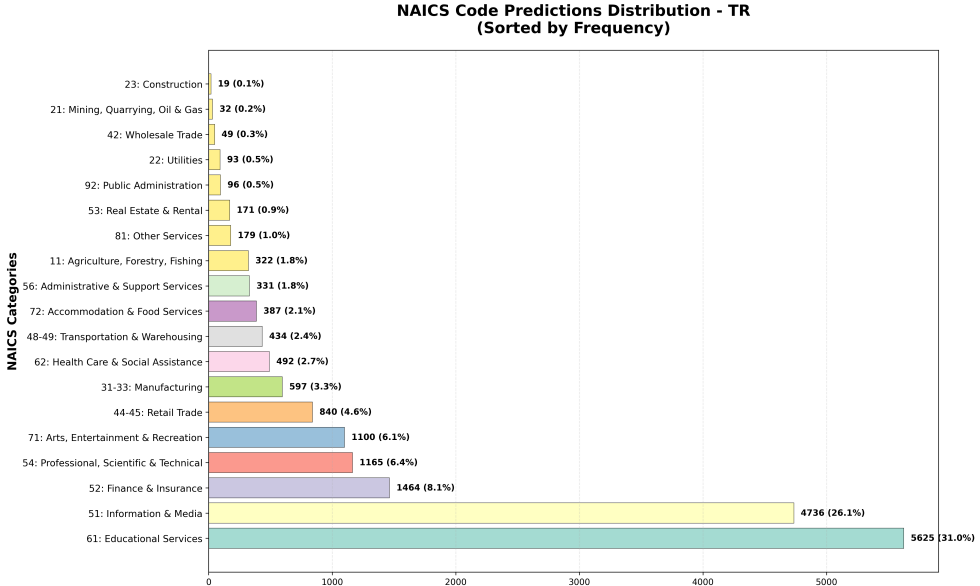
Step 3: Inference, High-Income Economies - South Korea



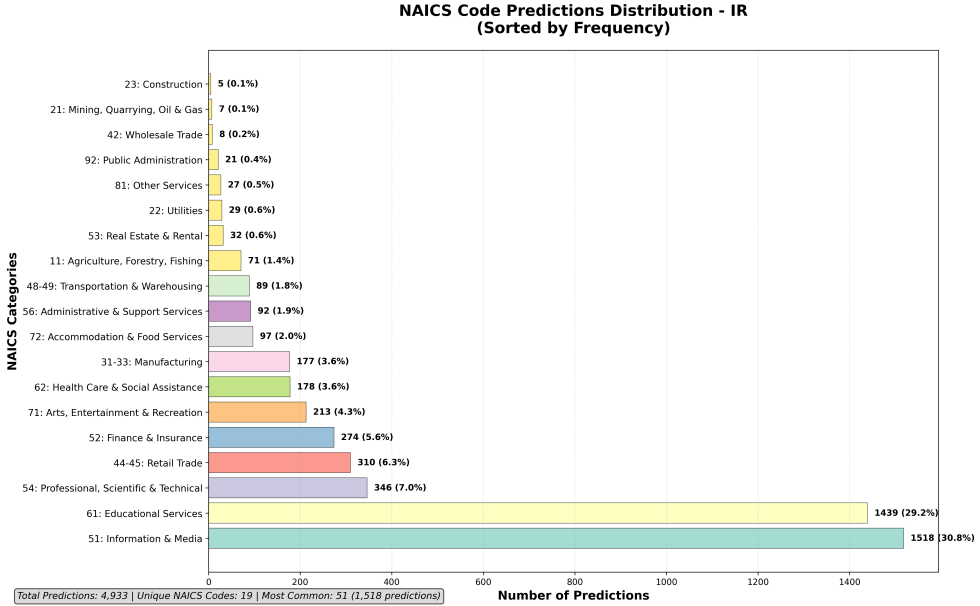
Step 3: Inference, Upper Middle-Income Economies - Brazil



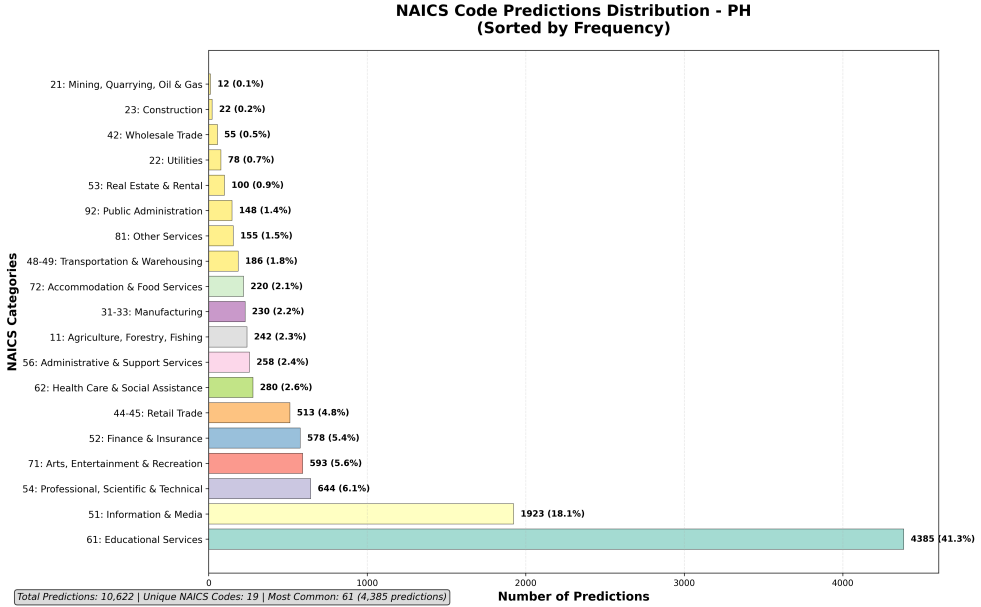
Step 3: Inference, Upper Middle-Income Economies - Turkey



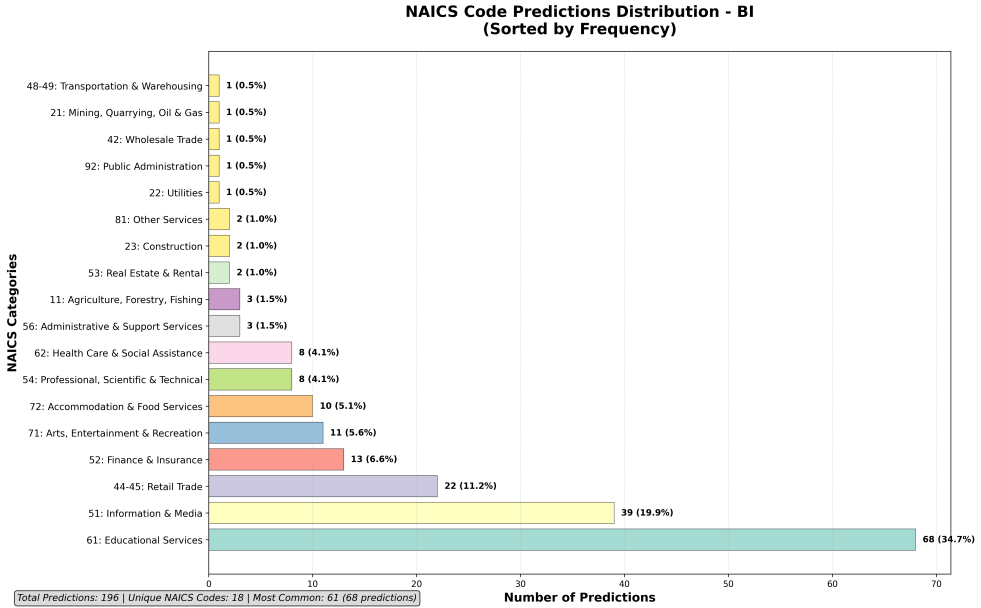
Step 3: Inference, Lower Middle-Income Economies - Iran



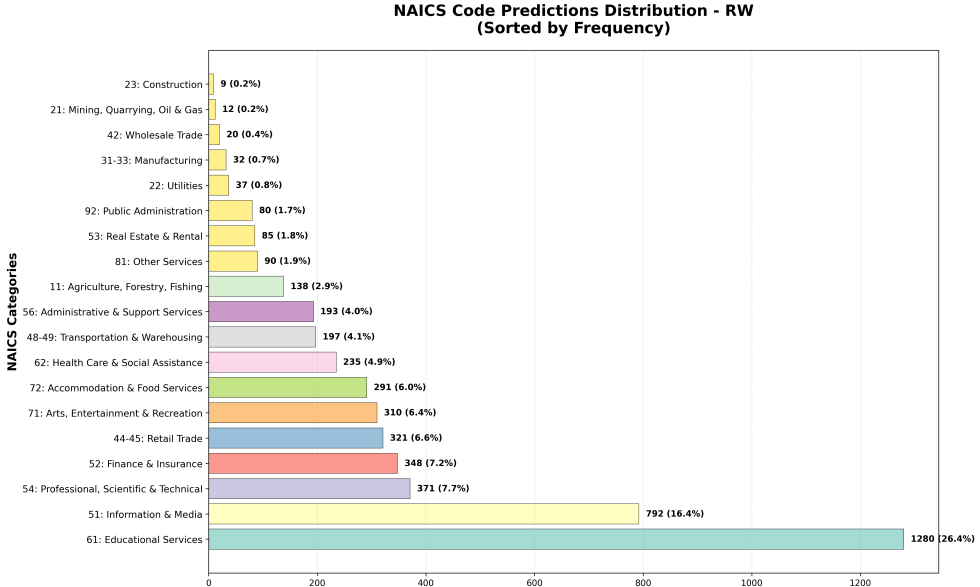
Step 3: Inference, Lower Middle-Income Economies - Philipines



Step 3: Inference, Low-Income Economies - Burundi



Step 3: Inference, Low-Income Economies - Rwanda



Conclusion: Broader Applications within GitHub

Key Takeaway: Our industry classification pipeline demonstrates how LLMs + semantic search can structure large-scale, unorganized repositories.

Potential Applications Across GitHub

- **Repository Discovery:** Improve search and recommendations by grouping repos by technical domain (e.g., AI, cybersecurity, cloud).
- **Ecosystem Mapping:** Identify growth areas in open source (e.g., emerging fields like quantum computing or climate tech).
- **Risk Compliance:** Detect repositories in regulated sectors (e.g., healthcare, finance) to assist with governance and audits.
- **Developer Insights:** Analyze contribution patterns by industry sector to understand trends in developer skills and expertise.
- **Enterprise Solutions:** Help organizations map internal projects to industry codes for better resource allocation and strategy.

Closing Note: This approach provides a scalable way to bring structure and insight to GitHub's massive open-source ecosystem.